

Note 8.3 Trigonometric Substitutions

Cases for Trigonometric Substitutions

- For $\sqrt{a^2 + x^2}$:
Let $x = a \tan \theta$, then $a^2 + x^2 = a^2 \sec^2 \theta$.
- For $\sqrt{a^2 - x^2}$:
Let $x = \sin \theta$ or $\cos \theta$, then $a^2 - x^2 = a^2 \cos^2 \theta$ ($a^2 \sin^2 \theta$).
- For $\sqrt{x^2 - a^2}$:
Let $x = \sec \theta$, then $x^2 - a^2 = a^2 \tan^2 \theta$.

Strategy

- Write down the substitution for x and then calculate the differential dx , then specify the domain of θ in the substitution.
- Substitute the trigonometric expression and the calculated differential into the integrand and then simplify the results algebraically.
- Integrate the trigonometric integral.
- Convert it back to the original variable x .

Exercises

- P470 43
- [P510 Question 5th](#)

Let $t = \sin a$, then

$$\int \frac{1}{t - \sqrt{1 - t^2}} dt = \int \frac{\cos a}{\sin a - \cos a} da$$

Then

$$\begin{aligned}
\int \frac{\cos a}{\sin a - \cos a} da &= \int \frac{\frac{1}{2}[(\cos a + \sin a) + (\cos a - \sin a)]}{\sin a - \cos a} da \\
&= \frac{1}{2} \int \frac{\cos a + \sin a}{\sin a - \cos a} da - \frac{1}{2} \int \frac{\sin a - \cos a}{\sin a - \cos a} da \\
&= \frac{1}{2} \int \frac{1}{\sin a - \cos a} d(\sin a - \cos a) - \frac{1}{2} a + C \\
&= \frac{1}{2} \ln |\sin a - \cos a| - \frac{1}{2} a + C \\
&= \frac{1}{2} \ln |t - \sqrt{1 - t^2}| - \frac{1}{2} \sin^{-1} t + C
\end{aligned}$$